

MATH7243-Machine Learning and Statistical Learning Theory

Instructor: He Wang

Class time and room: 3:00-4:30 pm | Ryder Hall 158

Office Hours:

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Some recommended textbooks:

1. An Introduction to Statistical Learning, with applications in R, by G. James, D. Witten, T. Hastie, R. Tibshirani. <https://www.statlearning.com/>
2. *The Elements of Statistical Learning: Data Mining, Inference, and Prediction* by Trevor Hastie, Robert Tibshirani, Jerome Friedman. <https://web.stanford.edu/~hastie/ElemStatLearn/>
3. *Hands-On Machine Learning with Scikit-Learn and TensorFlow: Concepts, Tools, and Techniques to Build Intelligent Systems* (2ed edition)- by Aurélien Géron
<https://github.com/ageron/handson-ml2>
4. *Pattern Recognition and Machine Learning* by Christopher Bishop
5. *Linear algebra and learning from data* by Gilbert Strang
6. *Machine Learning: A Probabilistic Perspective* by Kevin P. Murphy
7. *Understanding Machine Learning: From Theory to Algorithms* by Shai Shalev-Shwartz, Shai Ben-David
8. Other online sources will be provided on Canvas.
9. My recommended book list (include linear algebra and probability)
https://docs.google.com/spreadsheets/d/1j8LcyTxBTZ5Nh0-P1rb6NT2x_2iXFp1j8ptpvHYQWGA/edit?usp=sharing

Prerequisite: Basic knowledge (undergraduate level) about linear algebra, multivariable calculus, probability and statistics are required. Graduate level of applied linear algebra, probability1 is preferred.

Overview

Introduces both the mathematical theory of learning and the implementation of modern machine-learning algorithms appropriate for data science. Modeling everything from social organization to financial predictions, machine-learning algorithms allow us to discover information about complex systems, even when the underlying probability distributions are unknown. Algorithms discussed include regression, decision trees, clustering, and dimensionality reduction. Offers students an opportunity to learn the implications of the mathematical choices underpinning the use of each algorithm, how the results can be interpreted in actionable ways, and how to apply their knowledge through the analysis of a variety of data sets and models.

Practicum: Algorithms and Implementation:

The first portion of this course will focus on defining and applying the a variety of learning algorithms. We will start the class discussing the problem of learning a partially complete data labeling (Supervised Learning) and develop algorithms such as regression, support vector machines, and decision trees. We will then move to detecting patterns in unlabeled data using clustering and principle component analysis. By the end of the course, students should be familiar with the mathematical structure of these algorithms, and implications of any choices made in implementing them.

The machine learning is almost defined by its application. We will have frequent in class labs wherein we implement the algorithms we have discussed in the theory portion on real world data sets. In these labs, you will learn to use Python to obtain data, view it, clean it, analyze it, and

finally use machine learning algorithms to try to solve a problem. The labs are open ended, and students are expected to use their own ideas and intuition to try to get an honest, best fit on the data sets given. The lab portion will cumulate in a final project, where you (or possibly your team) will attempt a novel machine learning project within an area of interest.

Theory: Frameworks and Communication:

After we have a handle on high level methods of machine learning, we will develop we will use the PAC (partially approximately correct) model to define data, sampling, training, and the metrics we will use to evaluate a machine learning algorithms success. The PAC framework will allow us to discuss each component of the machine learning process, and the practical effects of all the relevant choices. It will also allow us to state and discuss the No Free Lunch Theorem (the bias/complexity trade off) about the limits of learnability. We will cover the VC dimension and the conditions under which we can guarantee an algorithm can learn a problem. Finally, we will analyze regularization, gradient decent and ensembling as extensions of the PAC model, and show how they can often lead to better results without violating the theorems above.

The final portion of the class is about the communication of machine learning results and ideas. As an applied mathematician, part of your job will to communicate your results to coworkers, bosses or investors that don't understand the math as well as you do. To this end, you will be expected to communicate clearly and effectively in both your labs and in your final project. The goal of your final project is to produce something you can put on a resume, a git-hub, or give as a presentation.

I will be available to help you throughout this process with writing, editing and communication. This course requires Python, Jupyter Notebook Server and github, all of which are free and open source.

Grade Breakdown

Written Homework (25%) - There will be three written assignments which will focus on theory.

Labs (30%) - There will be roughly 6 labs of which 4 must be completed. Labs will focus on the implementation of algorithms on real world data sets. Class time will be allotted for labs, but students may finish labs at home. In each lab, we will fit a real world data set using the algorithms of techniques introduced in that weeks' theory lecture. Labs will be graded out of 5 pts, 3 pts for completion, 2 pts for communication. There will also be a standing bonus of 2 pts for going above and beyond and exploring an interesting aspect of the parameter space, or getting a really good fit.

Midterm (15%)

Final Project (30%) - The final project will consist of a proposal (1 page), middle stage progress report(2-3 pages), project report (roughly 5 pages) and presentation (roughly 20 minutes with poster or slides). Project groups should contain 2-4 people.

Masters students: This class features an XN project with an industry partner. Masters students are encouraged to participate in this project. (<https://careers.northeastern.edu/experiential-network/>)

PhD students: If you would like to propose your own project, it can take one of three forms:

- A computational analysis of a data set using sufficiently complicated or novel techniques from this course.

- A theoretical presentation of a topic not covered in this course with a case study.
- Thesis or Lab project.

Excellent can consider to submit the poster to RISE(<https://www.northeastern.edu/rise/about/>)

I am more than happy to discuss possible projects in any of these categories with you.

The written component will be 75% of the grade, the presentation component 25%.

Rough project timeline: (Exact due day and time will be announced on Canvas.)

- Group Selection: January 27
- Project Proposal Deadline: February 20
- Progress Report: March 15
- First Draft: April 5
- Final Draft: April 12
- Presentations: April 13 - 24.

Machine Learning Algorithm Roadmap

- **Supervised** - Labeled training data

Regression - Predicting continuous values

Linear Regression

Nonlinear Regression and Functional Fitting

Radial Basis Functions

Neural Networks

Classification - Predicting discrete values

Logistic Regression

Support Vector Machines (SVM)

Decision Trees

Neural Networks

- **Unsupervised** - Unlabeled data, pattern detection and descriptive modeling

Principle Component Analysis

Clustering

Neural Networks

TDA

Further topics in Machine Learning (Not covered in this class)

Semi-Supervised - Image segmentation, sparse training data

Neural Networks (Regional-CNN, U-NET)

k-means Clustering

Graph Partitioning

Implementation

Model Ensembles

Online Learning

Large Data Sets

Distributed Computations

Natural Language Processing

Weeks		Topics	Assignment	Lab	Lab Due	Project
Week 1	Labor Day	Introduction to ML				
Week 2		Matrix Derivatives, Linear Regression		Lab		
Week 3		Ridge Regression, Lasso Regression, Gradient Decent	HW1 Due	Lab 1		
Week 4		Logistic Regression		Lab 2		
Week 5		LDA, QDA	HW 2 Due	Lab 3		
Week 6	Columbus Day /Indigenous Day	Resampling				Proposal Due
Week 7		Naïve Bayes, k-Nearest Neighbors				
Week 8		ANN, CNN +Lab	Midterm	Lab 4		
Week 9		RNN+ Lab				Check In
Week 10	Veterans'day	Support Vector Machines		Lab 5		
Week 11		PCA, Clustering	HW 3 Due			
Week 12	Thanksgiving	Clustering				Rough Draft
Week 13		Gaussian Mixtures		Lab 6		
Week 14		TDA, Presentation For Projects				Presentations